

Muhammad Mahad

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EDUCATION

City University of Hong Kong – Full Tuition Scholarship	Sep 2023 – Jun 2027
BSc. Computer Science, Cybersecurity Stream , CGPA: 3.93/4.3	
- Belt and Road Scholarship recipient – Hong Kong Government - Dean's List, Tiger Award (Top 10% of Students) and Elite 100 Student	

WORK EXPERIENCE

Data Analytics, AI, and System Integration Intern – Siemens Mobility Hong Kong	Jun 2025 – Present
- Built a two-hop bastion architecture for least-privilege data-center access (an SSH bastion host fronting a Kubernetes-hosted JumpServer PAM console, with per-asset local port forwarding), hardened Windows and Ubuntu servers against CIS benchmarks, and layered production web security with Fail2ban and HAProxy rate limiting. - Helped deploy a containerized, real-time microservice platform (Python, Spring Boot, Kafka, Redis, PostgreSQL) on Docker and Kubernetes for an intelligent transportation analytics use case (YOLOv8, NVIDIA DeepStream), building new pipeline features and later contributing C++ code to a performance-focused rewrite. - Built multi-stage GitLab CI/CD pipelines for parallel ARM and AMD builds, artifact management, and secure container deployment, improving deployment reliability.	
Digital Transformation and Automation Intern – Office of the Vice-President (RES), CityU	Jul 2026 – Present
- Architected a multi-agent AI system (Hermes agents on Docker): one master agent with admin privileges overseeing 7 isolated per-employee agents, each equipped with Honcho for persistent contextual memory, addressing AI governance, privilege-separation, and data-segregation risk. - Built AI-assisted automation workflows (n8n, Make.com, Dify) and set up secure jump host server access for staff.	
Team Lead, Technical Teaching Team – AWS Educate Cloud Ambassador, AWS Hong Kong	Dec 2025 – Present
Led a team of 9 to design and deliver 3 workshops on cloud fundamentals and AI/ML (including Amazon Bedrock, QuickSight) to 100+ attendees, coordinating with AWS professionals, faculty, and event and marketing teams, and distributing 37 certification vouchers.	
STEM Business Acceleration Intern – Youth Incubator Limited, Hong Kong	Jun 2024 – Aug 2024
- Analyzed a business problem (slow manual event-data collection) and designed a solution: scraped 700+ URLs via a Python and OpenAI API prompt-engineering pipeline into MongoDB, cutting collection time by 70%. - Engineered a secure login system (React, Twilio/Memberstack verification) that cut unauthorized access attempts by 40%; also built a client-facing WhatsApp chatbot (OpenAI API, Botpress, Node.js/Express) automating inquiries and registration, saving 20+ hours weekly.	

PERSONAL PROJECTS

Agentic AI for Cybersecurity (OpenClaw, LLMs, Docker, Kubernetes, React)	<i>Final Year Project</i>
- Architecting a hardened, multi-agent security platform (OpenClaw) in an isolated lab, with custom modules for reconnaissance, vulnerability scanning, log and threat correlation, and controlled exploit chaining, gated by a safety layer with MITRE ATT&CK-mapped checks and human approval for high-risk actions. - Evaluating agent performance against traditional tools and assessing AI/LLM security risks, including prompt injection and privilege escalation, then hardening the agent and re-testing to show which controls reduce risk.	
Security Research and Penetration Testing Write-Ups	muhammadmahad.netlify.app/blog
Published root-cause write-ups on complete exploitation chains: EternalBlue (MS17-010) to SYSTEM access; a Bolt CMS Twig SSTI to an NFS-leaked SSH key and GTFOBins privilege escalation; a file-upload RCE (CVE-2020-23828) with cron and credential-reuse escalation; and Jenkins Script Console RCE via default credentials.	

LEADERSHIP & INVOLVEMENT

• 28th of 150 teams , HKCERT Capture the Flag (Open Category) – HKCERT and DPO	<i>Dec 2025</i>
• PolyU x NuttyShell Cybersecurity CTF – Participant	<i>Mar 2026</i>
• Discipline Resident Tutor – Student Residence Office, CityU	<i>2025–2026</i>
• Teaching Assistant – CS2204: Fundamentals of Internet Applications Dev.	<i>SemA 2025 & 2026</i>

LANGUAGES & SKILLS

Languages: English (Fluent), Urdu (Fluent), Hindi (Verbal Fluency)
Technology Risk & Cybersecurity: Penetration Testing, Vulnerability Analysis, Application Security, AI/LLM Security, AI Governance, Risk Assessment, Identity and Access Management (JumpServer), System Hardening and Compliance (CIS Benchmarks, CIS-CAT), Intrusion Prevention (Fail2ban), Threat Correlation, MITRE ATT&CK
Cloud & Infrastructure: AWS, Docker, Podman, Kubernetes, Proxmox, Hyper-V, GitLab CI/CD, Linux
AI & Automation (Applied): Multi-Agent Workflows, Agentic AI (OpenClaw, Hermes), Generative AI (Amazon Bedrock), Prompt Engineering (OpenAI API, DeepSeek API), n8n, Make.com, Dify
Programming: Python, Java, JavaScript, TypeScript, C++, SQL
Soft Skills: Structured Problem-Solving, Technical Writing, Stakeholder Communication, Collaborative Leadership
Certifications: AWS Certified AI Practitioner (Responsible AI, RAG); NVIDIA – Fundamentals of Deep Learning (Jan 2026); HackTheBox Certified Penetration Testing Specialist (In Progress, 2026)